

Xuan Li Feng

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Research portfolio: xuan-lifeng.github.io

EDUCATION

MEng Biomedical Engineering

2023–2027 (expected)

Imperial College London

- Expected graduation: June 2027.
- Relevant coursework: Image Processing; Digital Biosignal Processing; Statistical Signal Processing & Inference; Biomedical Instrumentation.

RESEARCH EXPERIENCE

MEng Thesis: Anatomy-Aware Surgical Robotics

2026–2027

Imperial College London | Supervised by Dr Stuart Bowyer

- Developing a stereoscopic near-infrared fluorescence (NIRF) pipeline to reconstruct 3D vascular geometry from paired images for anatomy-aware surgical assistance.
- Current work focuses on vessel segmentation, stereo correspondence, triangulation, depth estimation, centreline extraction, and validation against synthetic ground truth and tissue-mimicking phantoms.
- Evaluating reconstruction reliability using depth and geometry error metrics; investigating failure modes from segmentation and correspondence uncertainty.

Deep Neural Receptive Fields in Mouse Auditory Cortex

2025–2026

Imperial College London | Supervised by Dr A. S. Kozlov

- Designed and trained a five-layer causal Conv1D model to predict auditory-cortical spiking responses from naturalistic ultrasonic vocalisations in mice.
- Used Jacobian-based automatic differentiation to derive dynamic spectrotemporal receptive fields and compare neural-network behaviour with STA, STC, MNE, and dSTRF analyses.

Research Internship: Magnetoelastic Energy Harvesting

Jun–Aug 2025

Zhejiang University | Principal Investigator: Professor Jikui Luo

- Conducted full-time research on GaIn-doped PVDF composites, using COMSOL multiphysics simulation and molecular-dynamics and density-functional-theory analysis to support experimental interpretation.

PROJECTS

Wearable Eye-Tracking AAC Device

2024–2025

Imperial College London

- Developed gaze and head-tilt input for a wearable AAC device using OpenCV, Raspberry Pi Zero 2 W, IMU sensing, and Google Glass; achieved 78.2% gaze-selection accuracy and 91.8% head-tilt accuracy with latency below 100 ms.

Remote Patient Monitoring System

2025

Imperial College London

- Developed components of a Java Swing clinical monitoring system with REST API and PostgreSQL backend, including real-time waveform display and digital-twin visualisation.

Biomedical Instrumentation Systems

2024–2026

Imperial College London

- Designed and validated multistage analogue cardiac and respiratory sensing systems using LM324 operational amplifiers, active filters, SPICE, and hardware prototyping; verified 3700 voltage gain and real-time monitoring at 30 breaths per minute.

PUBLICATION

Lu, J., Zhang, K., . . . , **Li Feng, X.**, . . . et al. (2025). *GaIn Induced Polarization Control in PVDF for High-Efficiency Energy Harvesting and Instantaneous Wireless Sensing*. Energy & Environmental Materials. [doi:10.1002/eem2.70201](https://doi.org/10.1002/eem2.70201)

TECHNICAL SKILLS

Medical Imaging & Vision	Stereo vision, 3D triangulation, image segmentation, centreline extraction, reconstruction validation, NIRF imaging concepts, OpenCV
Machine Learning & Neural Data	PyTorch, Conv1D, automatic differentiation/Jacobians, cross-validation, neural response prediction, model evaluation
Signal Processing & Statistics	Wiener and LMS filtering, AR modelling, spectral estimation, Cramér–Rao bounds, time-series analysis, hypothesis testing, Pearson correlation
Programming & Research Tools	Python, MATLAB, Java, JavaScript, Kotlin, Git/GitHub, Jupyter, PostgreSQL
Hardware & Systems	Embedded vision, Raspberry Pi, IMU sensing, PCB design, analogue filtering and amplification, SPICE, COMSOL Multiphysics